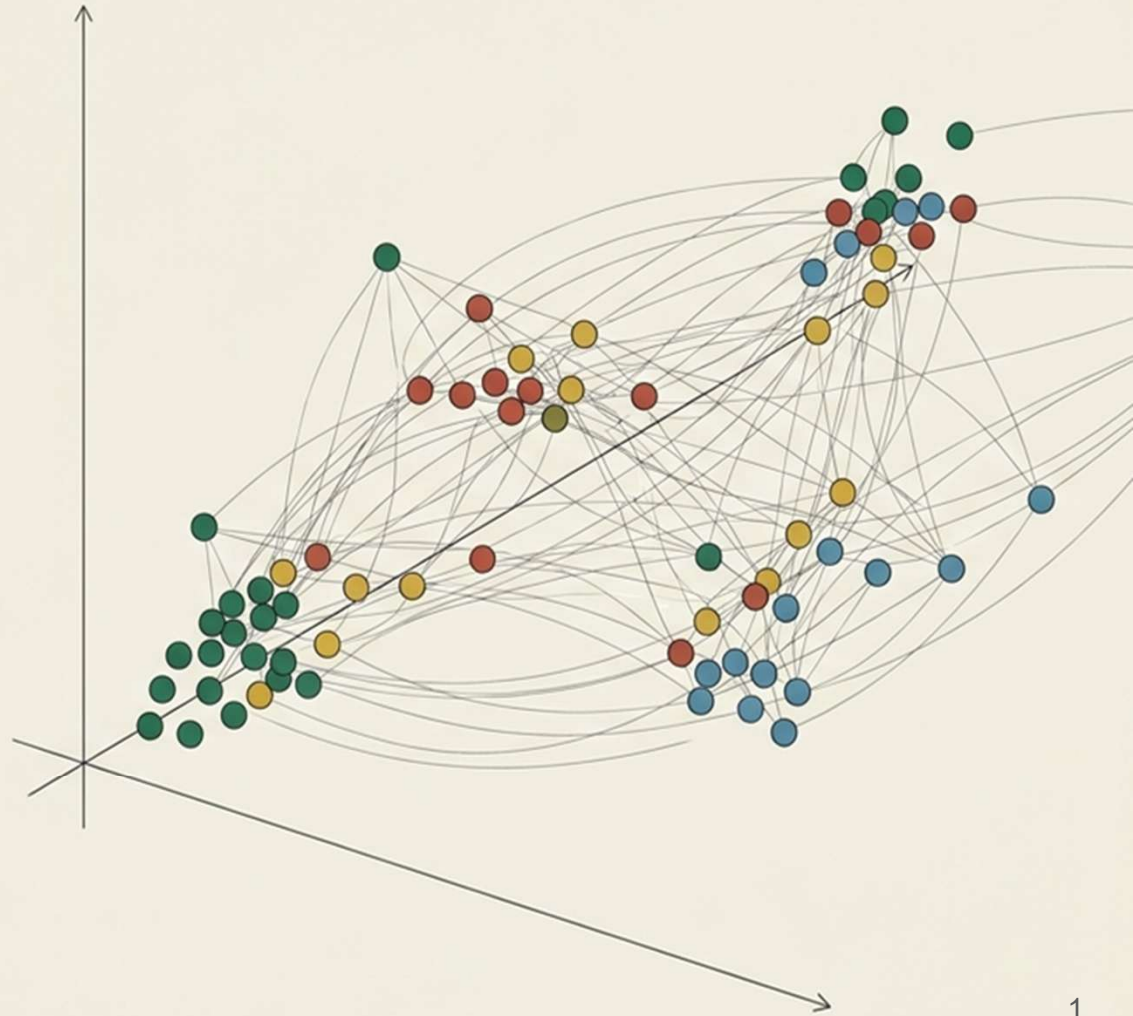


Integrating Cognition into Migration Models

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Existing Migration Studies vs. Empirical Reality



Traditional Models

Portraying human mobility through two contrasting lenses:

- Homogenous economic actors: Utility-maximizing individuals.
- Passive victims: Driven by external forces like conflict or disasters.



Empirical Reality

- Models fail to explain why **identical structural positions** lead to different decisions.
- Individuals may feel "trapped" despite a desire to move.
- Others stay in home communities despite economic incentives.



Moving Beyond Binaries

Migration is a multifaceted reality embedded in structural inequalities, cognitive landscapes, and social contexts.

Psychological factors such as attitudes, behavioral control, or social norms are seldom explicitly modeled.
— Klabunde and Willekens (2016)

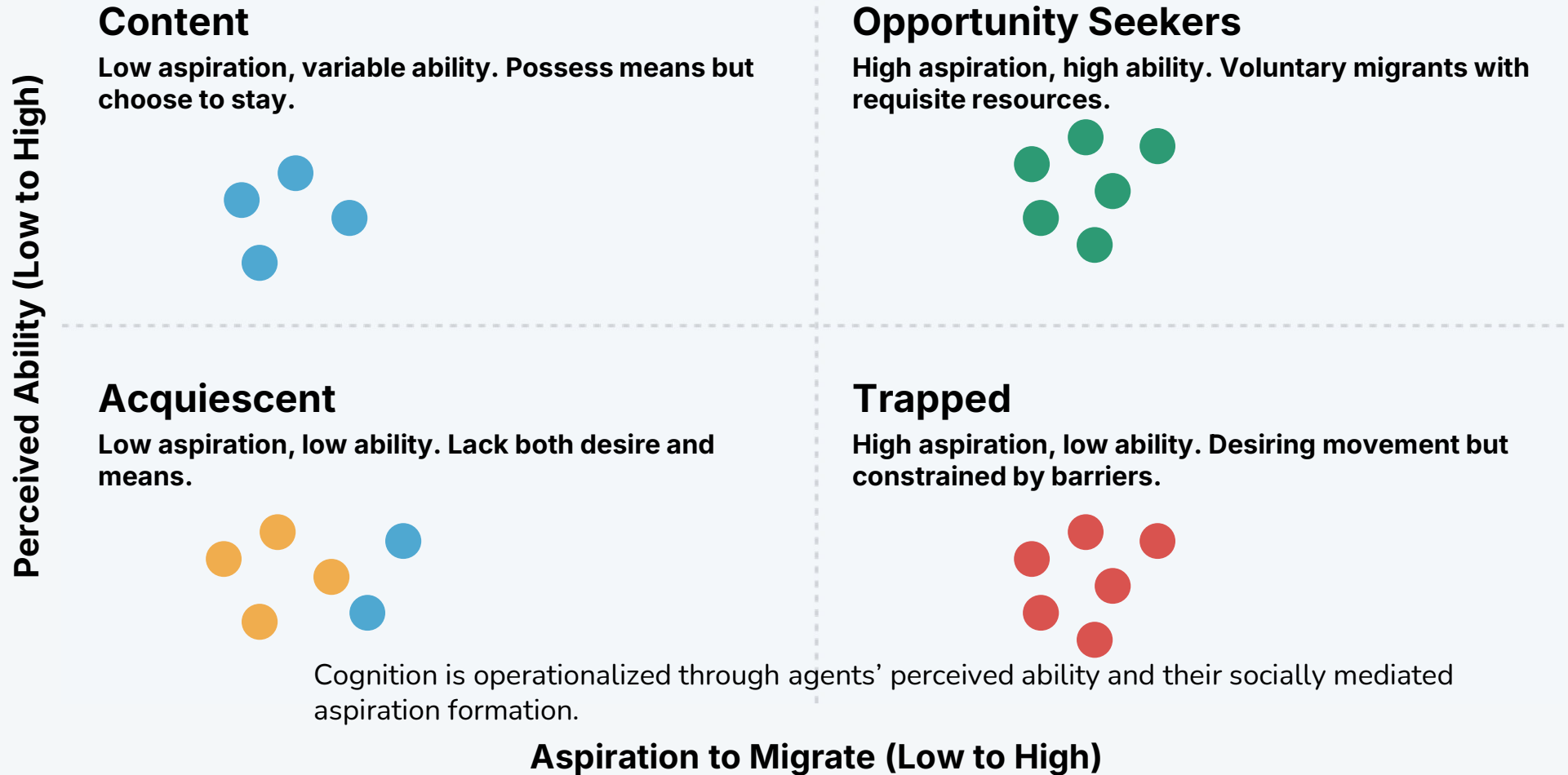
Structural Forces & Economics

Spatial Movement & Migration

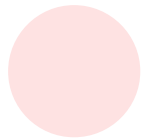
Carling & Schewel's (2018) Aspiration-Ability Framework:

Distinguishing the internal desire to migrate from the perceived external ability to do so.

Carling & Schewel 2018 Aspiration-Ability Framework

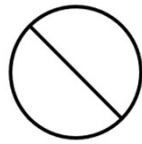


The Hierarchy of Aspiration: Safety-First Logic



Condition 1: Safety Threshold Not Met

Aspiration is 100% safety-driven. All other factors are muted until immediate survival is secured.



Condition 2: Safety Requirements Satisfied

The "Maslow Bottleneck" opens, allowing higher-level aspirations to form via a weighted blend.



70% **Internal Desires**
Economic, Family, Adventure

30% **Social Influence**
Neighbors & Global Diaspora

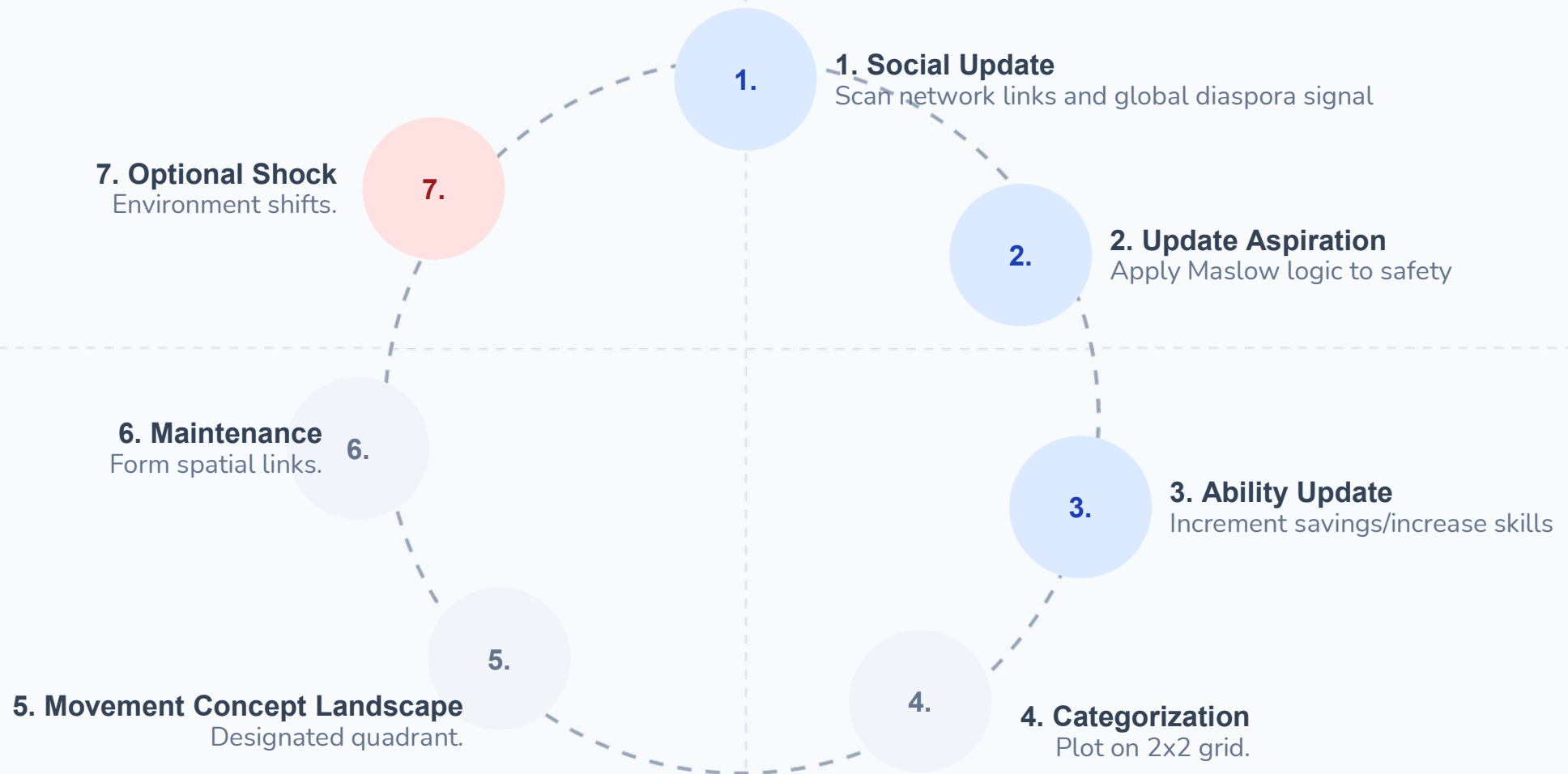
Model Logic: $\text{Aspiration} = \text{If}(\text{Safety} < \text{Threshold}, 1.0 * \text{Safety}, 0.7 * \text{Internal} + 0.3 * \text{Social})$

$$\text{Total Ability} = (0.2 \times \text{min}) + (0.8 \times \text{mean})$$



The Structural Reality: Modeling Ability. A compensatory system where significant bottlenecks—like a missing visa—heavily drag down the average, stranding an agent in the Low Ability threshold.

The Cognitive Engine Cycle



Simulation Scenarios & Migration Phenomena

Run 1: Culture of Migration

Primary Driver
Social Networks/Human Capital

Aspiration State
Emerging/Evolving

Ability State
Steady Growth

Gap (Asp. vs Ability)
Narrow & Manageable

Run 2: Economic Downturn

Primary Driver
External Disruption

Aspiration State
Spiked/Inflated

Ability State
Sharp Decline

Gap (Asp. vs Ability)
Wide & Frustrated

Run 3: High Aspiration

Primary Driver
Capability Accumulation

Aspiration State
Constant (High)

Ability State
Steady Growth

Gap (Asp. vs Ability)
Closing over time

Model Scale: 200 Individual Agents

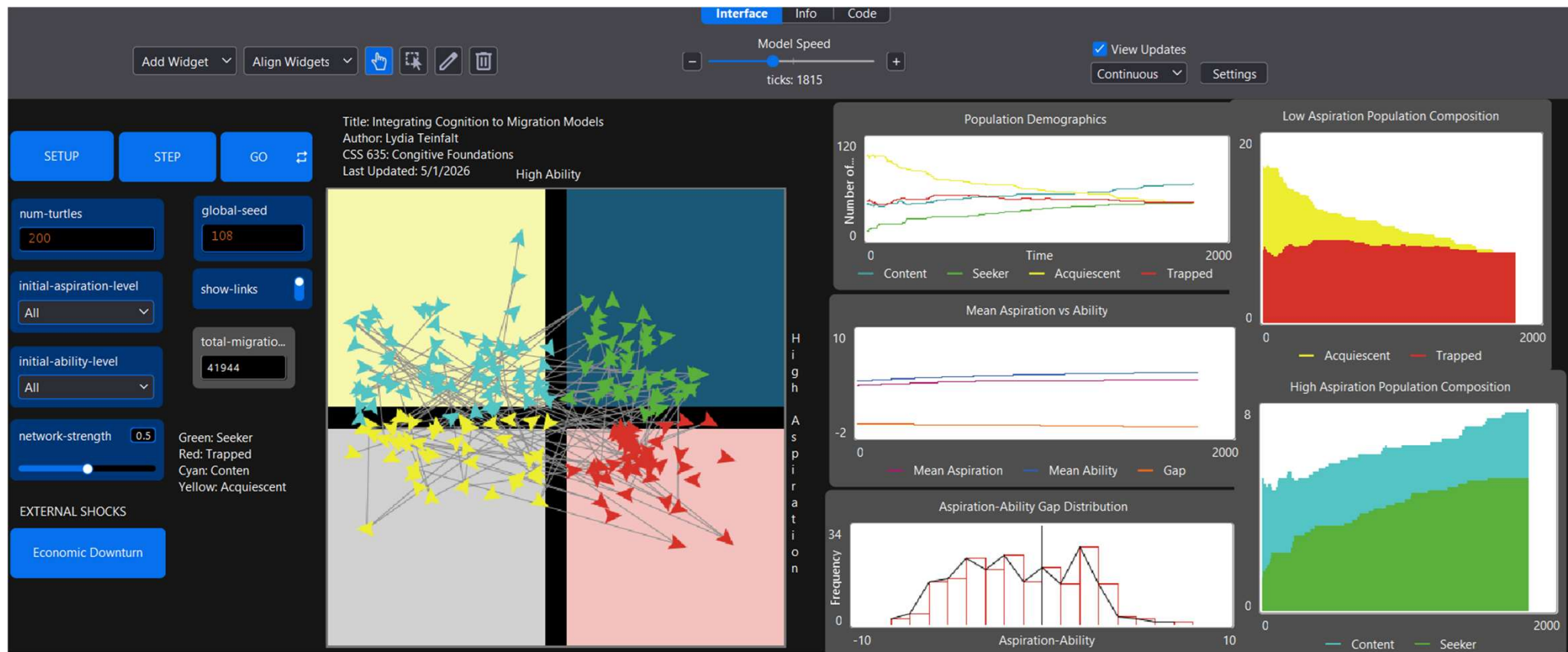
Duration: 1825 Ticks (1 tick = 1 day) = 5 years

Agents initialized with aspiration/ability (1-10). Ability grows via human capital simulation (learning/savings).

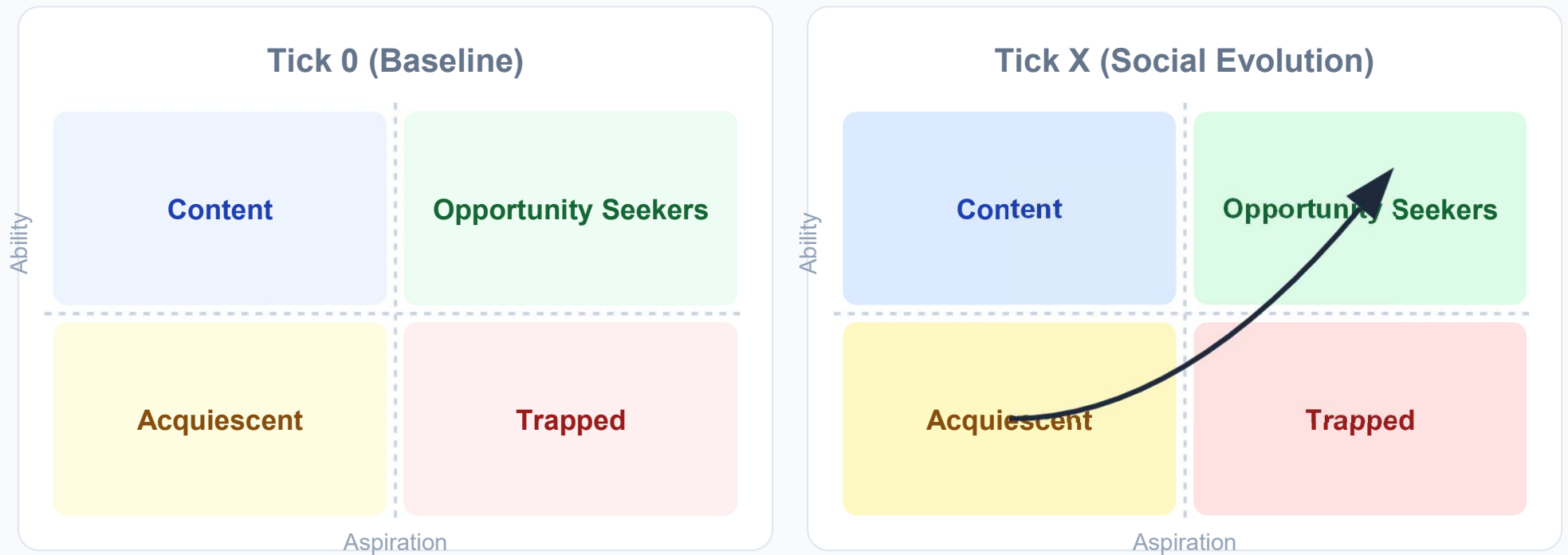
Aspiration increases through randomly assigned network connections.

Diaspora signal strength is calculated based on the density of opportunity seekers within the network.

NetLogo 7.03 Model

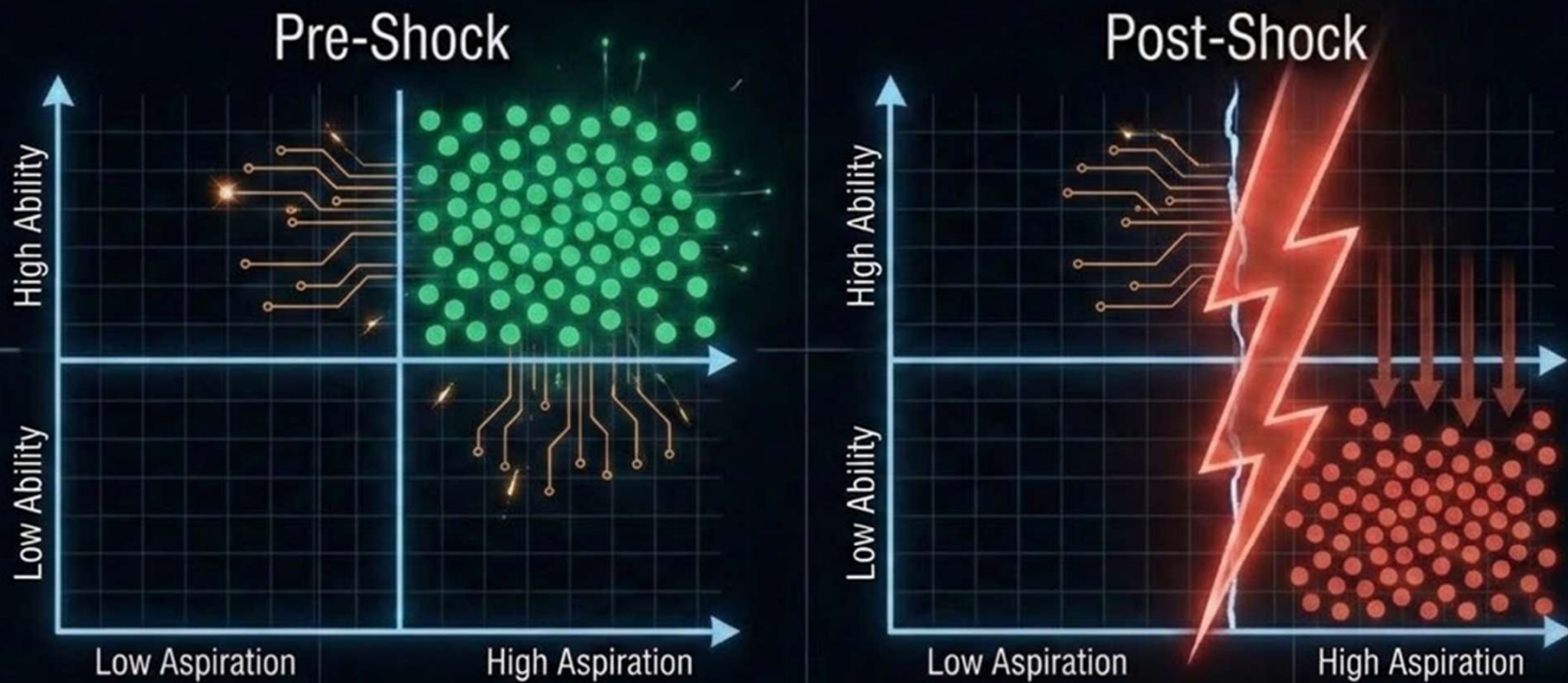


Baseline Simulation: The Culture of Migration



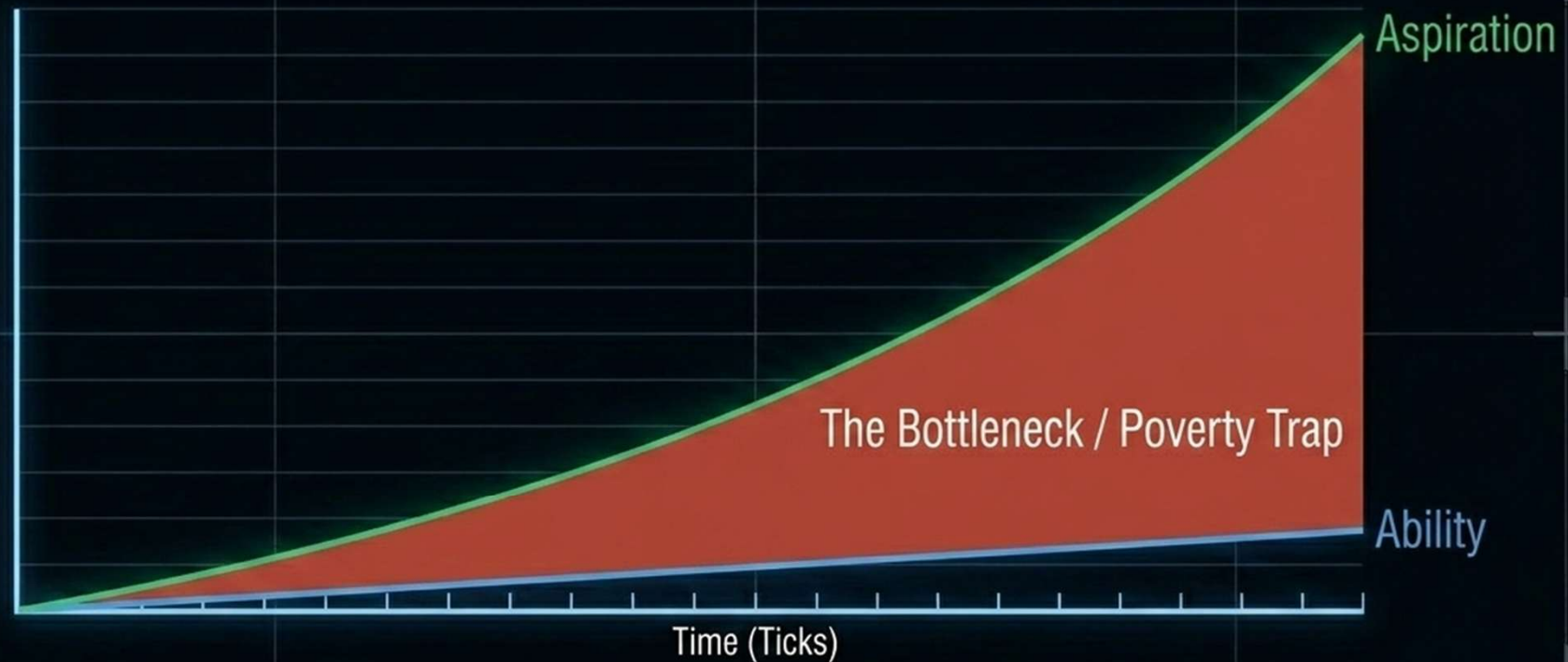
Insight: Natural social evolution. As the diaspora signal grows, the internal drive for those remaining increases. Aspiration is socially constructed and awakened by networks, regardless of immediate economic reality.

Simulation Shock: Creating Cognitive Misalignment



Insight: Systemic failures don't just stop movement—they strip away ability without erasing desire. Opportunity Seekers are instantly transformed into Trapped populations, creating a structural precursor to irregular migration.

Blocked Incremental Actualization: The Poverty Trap



Blocked Incremental Actualization. While ability grows linearly through savings, aspirations compound socially. The inescapable gap creates a permanent cognitive bottleneck for populations in precarious conditions.

The background of the slide features a dark navy blue field densely populated with small, bright orange dots. These dots are scattered across the entire frame, with a slightly higher concentration towards the top and bottom edges. A faint, light gray grid is overlaid on the background, consisting of both horizontal and vertical lines that create a subtle pattern of squares.

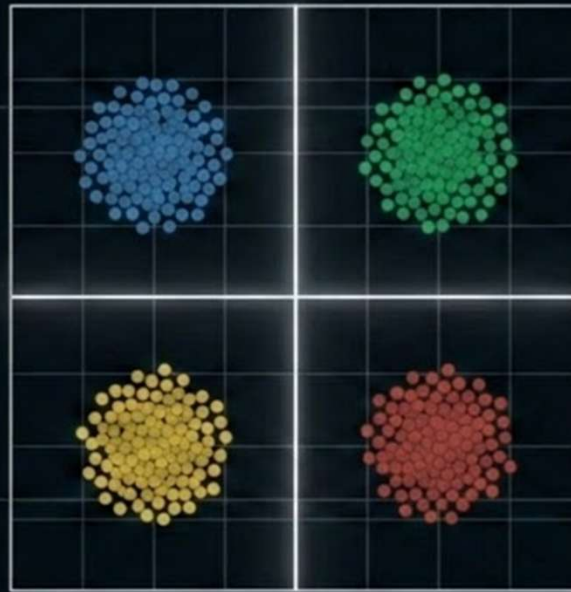
MIGRATION IS NOT A SIMPLE RESPONSE TO POVERTY.

Poverty is frequently a barrier to migration, not a driver. The most dangerous systemic state—the “Cognitive Gap”—occurs when aspirations remain high but structural ability is stripped away, leaving a frustrated population indefinitely trapped.

Scope & Computational Limitations

The International Focus:
Distance, information
asymmetries, and hard
structural barriers (like
visas and language)
uniquely magnify the gap
between aspiration and
ability.

Quantifying the Subjective:
A fundamental limitation of
Agent-Based Models is the
necessary
oversimplification of
treating subjective human
emotions as rigid Cartesian
coordinates.



By integrating cognition, Agent-Based Models cease to be mere economic calculators. They become engines for empathy, capable of simulating the complex psychological state of global migration.

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<https://github.com/lydiateinfalt/Agent-Based-Modeling-Migration-and-Cognition>

Selected images from slides were created with the NotebookLM and Advanced Gemini